

AI-Powered Resume Optimization Using a Chained Prompt Workflow

Goal

To transform a candidate's existing resume into a targeted, ATS-friendly resume for a Data Analyst Intern position using a three-step chained AI workflow: analysis, resume generation, and self-critique with final refinement.

Prompt 1

Act as an expert technical recruiter and ATS resume specialist. I will provide a candidate's raw resume and a target Data Analyst Intern job description. Your task is to analyze both and create a structured resume improvement plan. Identify:

1. The top 8 skills from the job description that should be emphasized.
2. Which of those skills are already supported by the candidate's resume.
3. The main gaps or weaknesses in the current resume.
4. Which projects or experiences are most relevant to the target role.
5. Specific improvements that should be made to the professional summary, skills section, and project descriptions.
6. A recommended structure for an ATS-friendly one-page resume.

Do not rewrite the resume yet. Only analyze the information and create the improvement plan. Candidate Resume: [PASTE THE RAW RESUME HERE] Target Job Description: Data Analyst Intern We are looking for a Data Analyst Intern to support business teams with data-driven insights. Responsibilities:

- Clean and analyze structured datasets.
- Write SQL queries to extract business information.
- Create dashboards and reports using Power BI or similar tools.
- Use Python for data analysis and automation.
- Identify trends and anomalies in business data.
- Communicate insights clearly to stakeholders. Requirements:
- Strong SQL knowledge.
- Python and Pandas experience.
- Knowledge of Power BI or Tableau.
- Understanding of data cleaning and visualization.
- Analytical and problem-solving skills.
- Good communication skills.

AI Output:

Resume Improvement Plan: Data Analyst Intern

1. Top 8 Skills to Emphasize (from JD)

1. SQL (querying/business data extraction)
2. Python for data analysis & automation
3. Pandas (data manipulation)
4. Power BI / Tableau (dashboards & reporting)
5. Data cleaning
6. Data visualization
7. Analytical/problem-solving skills (trend & anomaly detection)
8. Communication skills (translating insights to stakeholders)

2. Already Supported by Current Resume

Skill	Evidence	Strength
SQL	Listed under Programming; SQL Certification (CodeChef)	Weak — no project shows SQL in action
Python	Listed; used in ETL pipeline project	Moderate
Pandas	Listed under Data Analysis	Weak — not tied to a specific outcome
Power BI	Listed under Data Visualization	Weak — not used in any listed project
Data cleaning	Listed explicitly	Weak — no example given
Data visualization	Streamlit dashboard project; Matplotlib/Plotly/Tableau listed	Moderate
Analytical/problem-solving	Anomaly detection project (outlier detection)	Moderate-strong, but ML-flavored rather than business-analytics-flavored
Communication	Listed as soft skill only	Weak — no demonstrated evidence

3. Main Gaps/Weaknesses

- **No SQL project evidence.** SQL is the #1 JD requirement, but the resume only shows a certification — no query work, database, or reporting task tied to it.
- **No Power BI/Tableau project evidence.** Both tools are listed as skills but never appear in a project bullet. This is a major credibility gap for an ATS and a human reviewer.
- **Projects lean ML/engineering, not business analytics.** The two projects (retail ETL system, anomaly detection ML model) read as software/data-engineering work, not "business insight" work. Neither has quantified business impact (e.g., cost saved, stock-outs reduced, accuracy %).
- **No metrics/quantification.** Bullets describe *what was built*, not *what it achieved* (e.g., "reduced latency" but no baseline; "identified stock-out risks" but no number of risks flagged or accuracy).
- **Summary is generic.** "Aspiring Data Analyst... skilled in..." reads as a skills list restated in prose rather than a value proposition tailored to a business-analytics intern role.

- **Communication skill is unsubstantiated.** No mention of presenting findings, writing reports, or working with stakeholders/teams.
- **Certifications section is a mix of relevant and low-signal items.** The Deloitte Data Analytics Job Simulation is strong and relevant; the "Repo-Ready" GitHub workshop is not relevant to this JD and dilutes the section.
- **No explicit "Data Cleaning" project callout**, even though it's a named skill — recruiters/ATS look for it demonstrated, not just listed.

4. Most Relevant Projects/Experience (ranked)

1. **Real-Time Retail Analytics Suite** — most relevant. Has ETL (data cleaning/processing), dashboarding (Streamlit), and business KPIs (stock-out risk). Needs reframing toward business insight/SQL rather than backend engineering.
2. **AI-Driven Anomaly Detection for Defence Supply Chain** — moderately relevant. "Anomaly detection" maps directly to JD's "identify trends and anomalies," but currently framed as an ML hackathon project rather than a business-analytics task. Keep but reframe the language away from heavy ML jargon.
3. **Deloitte Data Analytics Job Simulation (Forage)** — highly relevant credential; currently under-leveraged as just a certification bullet. Could be elevated with a one-line description of what was done in the simulation (if it involved SQL/Excel/dashboards).

5. Specific Section-Level Improvements

Professional Summary

- Rewrite to lead with the JD's language: SQL, Python/Pandas, dashboarding, business insight generation — in that priority order.
- Replace generic "aspiring" framing with a value statement (what the candidate can *do* for a team), not just a list of tools already stated in the Skills section (avoid redundancy).
- Add a quantifiable or concrete anchor (e.g., number of projects, dashboards built, or dataset scale handled).

Skills Section

- Reorder to match JD priority: SQL → Python/Pandas → Power BI/Tableau → Excel → Data Cleaning → Visualization (Matplotlib/Plotly) → Streamlit → Basic ML.
- Consider splitting into JD-mirrored categories (e.g., "Querying & Databases," "Analysis & Automation," "Visualization & BI Tools") so an ATS keyword scan and a human skim both hit SQL/Power BI immediately, not buried mid-list.
- Drop or de-emphasize "Basic Machine Learning" — it's not in the JD and signals a different specialization.

Project Descriptions

- Retail Analytics project: add/emphasize any SQL used for querying the MongoDB/transaction data (even if minimal), and reframe bullets around *business outcomes* (e.g., "flagged X potential stock-outs," "reduced manual reporting time by X%") rather than pure system-building.

- Anomaly Detection project: reframe first bullet to lead with the business question ("analyzed supply chain data to detect anomalies indicating fraud/delay risk") before the technical method, and add a concrete result (e.g., number/percentage of anomalies correctly flagged).
- Consider adding a **third, smaller project or bullet specifically demonstrating SQL + Power BI together** if the candidate has any coursework, simulation, or personal project that used them — this directly closes the biggest gap. If none exists, this should be flagged as a priority to build before applying.

6. Recommended ATS-Friendly One-Page Structure

1. **Header** — Name, location, phone, email, LinkedIn, GitHub (current format is fine)
2. **Professional Summary** — 3 lines, JD-keyword-led (SQL, Python, Power BI, business insights)
3. **Technical Skills** — grouped by JD-mirrored categories, SQL and Power BI visible near the top
4. **Projects** — 2–3 projects, each with 2–3 bullets in "Action verb + tool + business outcome/metric" format
5. **Certifications** — keep only role-relevant ones (Deloitte simulation, SQL cert, Python cert); drop or move unrelated ones (e.g., GitHub workshop) to a minor "Additional" line or remove
6. **Education** — keep as-is, near bottom for internship-level resumes (skills/projects should lead since the candidate is still a student)

Standard single-column layout, consistent bullet formatting, no tables/graphics/text boxes — keeps it ATS-parseable.

Prompt 2

Using only the resume improvement plan you just created, rewrite the candidate's resume for the Data Analyst Intern position. Apply the recommendations from your previous analysis. Requirements:

- Create an ATS-friendly one-page resume.
- Prioritize skills that match the target job description.
- Write a concise professional summary.
- Organize technical skills into clear categories.
- Rewrite the project descriptions using strong action verbs.
- Emphasize Python, SQL, Pandas, Power BI, data cleaning, data analysis, and visualization where supported by the original resume.
- Do not invent companies, internships, certifications, achievements, technologies, responsibilities, or numerical results that were not provided.
- Keep the language professional and concise.
- Remove unnecessary information.
- Make the projects relevant to a Data Analyst role.
- Do not include tables, graphics, icons, or columns.
- Produce only the revised resume content.

Use the previous analysis and improvement plan as the basis for every change.

AI Output

DONTHRAMONI SIDDARTH

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PROFESSIONAL SUMMARY

Data Science undergraduate pursuing a Data Analyst Intern role, with hands-on experience in SQL, Python, and Pandas for data cleaning and analysis, and in Power BI, Tableau, and Streamlit for building dashboards and reports. Experienced in developing ETL pipelines and analyzing structured data to identify trends and anomalies. Strong analytical, problem-solving, and communication skills suited to a business-facing analyst role.

TECHNICAL SKILLS

Querying & Databases: SQL

Analysis & Automation: Python, Pandas, NumPy, Data Cleaning

Visualization & BI Tools: Power BI, Tableau, Excel, Matplotlib, Plotly, Streamlit

Tools: GitHub

Soft Skills: Analytical Thinking, Problem-Solving, Communication, Adaptability, Teamwork

PROJECTS

Real-Time Retail Analytics Suite – Srinidhi Vegetables

- Analyzed live transaction and inventory data from a simulated retail environment to support business decision-making.
- Developed a Python ETL pipeline to extract, clean, and process inventory data with less than 60-second latency.
- Built a Streamlit dashboard to visualize retail KPIs, enabling identification of potential stock-out risks.
- Built the underlying data-generation system using Node.js and MongoDB to simulate real-time retail transactions.

AI-Driven Anomaly Detection for Defence Supply Chain

- Analyzed logistics and supply chain data during a university-level hackathon to detect anomalies and irregular patterns.
- Applied outlier detection techniques to identify unusual patterns associated with fraud risk and supply chain delays.

CERTIFICATIONS

- Data Analytics Job Simulation – Deloitte (Forage), August 2025
- SQL Certification – CodeChef
- Python Programming – CodeTantra

EDUCATION

Kalasalingam Academy of Research and Education, Madurai

B.Tech – Computer Science & Engineering (Data Science), 2023–2027 (Expected) | CGPA: 7.90

Narayana College, Hyderabad

Intermediate (MPC), 2021–2023 | Score: 89.9%

Prompt 3

Using only the resume improvement plan you just created, rewrite the candidate's resume for the Data Analyst Intern position. Apply the recommendations from your previous analysis. Requirements:

- Create an ATS-friendly one-page resume.
- Prioritize skills that match the target job description.
- Write a concise professional summary.
- Organize technical skills into clear categories.
- Rewrite the project descriptions using strong action verbs.
- Emphasize Python, SQL, Pandas, Power BI, data cleaning, data analysis, and visualization where supported by the original resume.
- Do not invent companies, internships, certifications, achievements, technologies, responsibilities, or numerical results that were not provided.

- Keep the language professional and concise.
- Remove unnecessary information.
- Make the projects relevant to a Data Analyst role.
- Do not include tables, graphics, icons, or columns.
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AI Output:

Critique Table

Criterion	PASS/ FAIL	Reason	Correction Needed
1. Clearly targets Data Analyst Intern role	PASS	Skills ordering, project framing, and certifications all point toward data analytics work.	None — reinforced by Correction #3 below for consistency.
2. JD skills prominently represented where supported	PASS	SQL and Power BI are listed first in their categories, matching JD priority. Note: neither is demonstrated in a project — this reflects a real gap in the original resume, not a drafting error, so nothing can be added without fabricating.	None (inherent source-material limitation).
3. Professional summary matches target position	FAIL	The summary opens with "Data Science undergraduate" and never names the target role. A recruiter scanning the top of the page has no immediate confirmation this is a Data Analyst Intern application.	Add explicit "pursuing a Data Analyst Intern role" framing to the opening sentence.
4. Project descriptions relevant to data analysis	FAIL	The first bullet of the Retail Analytics project leads with "Node.js and MongoDB" — backend/engineering tools that are not part of the JD and dilute the analytics framing right at the top of the project.	Move the Node.js/MongoDB detail out of the lead bullet; lead with the analysis outcome instead.
5. Python, SQL, Pandas, Power BI, data cleaning, visualization emphasized where supported	PASS	All six appear in the Skills section in JD-priority order; Python/Pandas/data cleaning are additionally reinforced in the ETL project bullet.	None.

Criterion	PASS/ FAIL	Reason	Correction Needed
6. No unsupported experience/technology/achievement/numeric claim introduced	FAIL	The summary states the candidate has "the communication skills to translate data into clear, actionable insights for stakeholders" — this implies a demonstrated outcome, but the original resume only lists "Communication" as an unproven soft skill. This is an overclaim beyond the source material.	Reword to state possession of the skill without implying a demonstrated stakeholder-facing achievement.
7. Concise enough for one-page ATS resume	PASS	Content fits comfortably on one page with room to spare; no trimming needed.	None.
8. ATS-friendly formatting (no tables/columns/graphics/icons)	PASS	Single-column layout, standard bullets, no tables, graphics, icons, or text boxes.	None.
9. Project descriptions use clear action verbs	PASS	Bullets consistently open with strong verbs (Analyzed, Developed, Built, Applied).	None.
10. Avoids unnecessary/irrelevant information	PASS (with minor note tied to #4)	Irrelevant certification (GitHub workshop) and low-value SSC line were already removed in the prior revision.	Addressed via Correction #4 above.

Summary: 7 PASS / 3 FAIL. Corrections needed: (1) sharpen summary's role targeting, (2) reframe lead project bullet away from engineering-tool emphasis, (3) soften the unsupported communication claim.

Applying these corrections now — no new facts, tools, or numbers are being added, only rewording and reordering of existing, source-supported content.